

GlucoConnect Clinical Rule Engine

Overview

The GlucoConnect Clinical Rule Engine is a deterministic clinical decision-support system designed to analyze patient glucose readings, historical trends, medication adherence, meal patterns, and lifestyle factors to generate personalized risk assessments and recommendations.

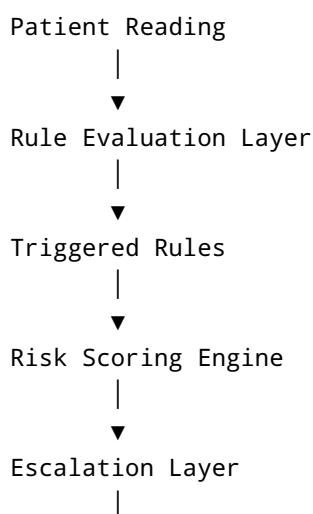
Unlike traditional threshold-based monitoring systems, the engine incorporates trend analysis, glycemic variability detection, recurrence monitoring, and patient-specific personalization to provide proactive diabetes management support.

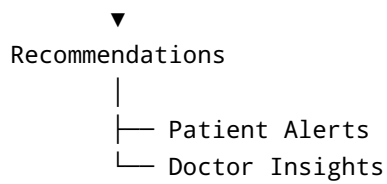
My Contribution

I designed and implemented the complete Clinical Rule Engine layer, including:

- Clinical rule evaluation framework
 - Risk stratification engine
 - Diabetes-type personalization
 - Medication adherence analysis
 - Trend velocity detection
 - Glycemic variability analysis
 - Meal intelligence scoring
 - Recurrence detection logic
 - Escalation and prioritization layer
 - Rule confidence scoring system
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System Architecture





Rule Categories

Critical Risk Rules

CRITICAL_HYPOGLYCEMIA

Detects glucose readings below 70 mg/dL.

Purpose:

- Immediate hypoglycemia detection
- Prevent severe clinical events

CRITICAL_HYPERGLYCEMIA

Detects glucose readings above 300 mg/dL.

Purpose:

- Hyperglycemic crisis detection
- DKA prevention support

Trend Analysis Rules

FASTING_HIGH_TREND

Triggers when:

- Last 3 fasting readings exceed patient-specific thresholds

Purpose:

- Detect persistent poor glucose control
 - Identify long-term deterioration
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RAPID_GLUCOSE_RISE

Measures glucose increase velocity using timestamps.

Purpose:

- Detect emerging hyperglycemia before critical levels are reached
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RAPID_GLUCOSE_DROP

Measures glucose decline velocity.

Purpose:

- Detect potential hypoglycemia before dangerous thresholds occur
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Diabetes-Type Personalization

The engine supports:

- Type 1 Diabetes
- Type 2 Diabetes
- Gestational Diabetes

Each patient type uses individualized targets.

Example:

Diabetes Type	Fasting Max	Post Meal Max
Type 1	130	180
Type 2	126	180
Gestational	95	140

This ensures clinically relevant recommendations rather than one-size-fits-all thresholds.

Glycemic Variability Analysis

GLUCOSE_VARIABILITY_RISK

The system calculates:

1. Mean Glucose
2. Standard Deviation

3. Coefficient of Variation (CV)

Formula:

$CV = \text{Standard Deviation} / \text{Mean}$

Rule Trigger:

$CV > 36\%$

Clinical Significance:

High glycemic variability is associated with:

- Increased hypoglycemia risk
 - Increased hyperglycemia risk
 - Poor glucose stability
 - Higher complication rates
-

Medication Adherence Intelligence

MISSED_MEDICATION_RISK

Triggers when:

- Glucose > 180 mg/dL
- Three or more missed medication events are detected within recent logs

Purpose:

- Detect treatment non-adherence patterns
 - Improve physician visibility
-

Meal Intelligence Engine

HIGH_GLYCEMIC_LOAD_MEAL

Analyzes meal descriptions using weighted glycemic scoring.

Examples:

High Impact Foods:

- White rice
- Jalebi
- Gulab Jamun

- Paratha

Protective Foods:

- Salad
- Dal
- Paneer

Purpose:

- Explain post-meal spikes
 - Improve nutritional recommendations
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Recurrence Detection

RECURRENT_HYPOGLYCEMIA

Triggers when:

- Multiple hypoglycemic episodes occur within recent history

Purpose:

- Detect repeated dangerous events
 - Recommend physician review
-

RECURRENT_POST_MEAL_SPIKES

Triggers when:

- Multiple consecutive post-meal spikes are detected

Purpose:

- Identify persistent dietary or medication issues
-

Risk Scoring Engine

Each triggered rule contributes to an overall patient risk score.

Severity	Score
Critical	100
High	40

Severity	Score
Medium	20
Low	10

Risk scores are capped at 100.

Risk Levels:

Score	Level
0-19	LOW
20-39	MEDIUM
40-99	HIGH
100	CRITICAL

Escalation Layer

Triggered risks are automatically prioritized.

Output:

```
{
  "primaryRisk": "RECURRENT_HYPOGLYCEMIA",
  "secondaryRisks": [
    "RAPID_GLUCOSE_DROP",
    "GLUCOSE_VARIABILITY_RISK"
  ]
}
```

Purpose:

- Highlight the most clinically significant issue first
 - Reduce alert fatigue
-

Rule Confidence Scoring

The engine generates a deterministic Rule Confidence Score.

Factors:

- Historical data availability

- Pattern consistency
- Clinical signal strength

Example:

```
{
  "rule": "FASTING_HIGH_TREND",
  "ruleConfidenceScore": 0.94
}
```

This score reflects confidence in the rule outcome and is not generated using machine learning.

Output Format

```
{
  "riskScore": 82,
  "riskLevel": "HIGH",
  "primaryRisk": "RECURRENT_POST_MEAL_SPIKES",
  "secondaryRisks": [
    "HIGH_GLYCEMIC_LOAD_MEAL"
  ],
  "triggeredRules": [
    {
      "rule": "RECURRENT_POST_MEAL_SPIKES",
      "severity": "high",
      "ruleConfidenceScore": 0.96
    }
  ],
  "alerts": [
    "Your blood sugar has spiked after your last few meals."
  ]
}
```

Impact

The Rule Engine transforms GlucoConnect from a reactive glucose monitoring application into a proactive clinical decision-support platform.

Key Outcomes:

- Personalized recommendations
- Early risk detection
- Pattern-based analysis
- Medication adherence monitoring

- Improved physician visibility
- Reduced likelihood of adverse diabetic events

This architecture prioritizes explainability, clinical transparency, and patient safety while remaining lightweight enough for real-time deployment.